

Output Port Processing and Queueing in Routers

1 Output Port Processing

Output port processing is responsible for taking packets stored in the output port's memory (buffer) and transmitting them onto the outgoing link. The main operations performed at the output port include:

- **Packet scheduling:** selecting which packet should be transmitted next.
- **Dequeuing:** removing the selected packet from the buffer.
- **Link-layer processing:** encapsulation, trailer addition, and protocol-specific functions.
- **Physical-layer transmission:** sending bits onto the physical medium.

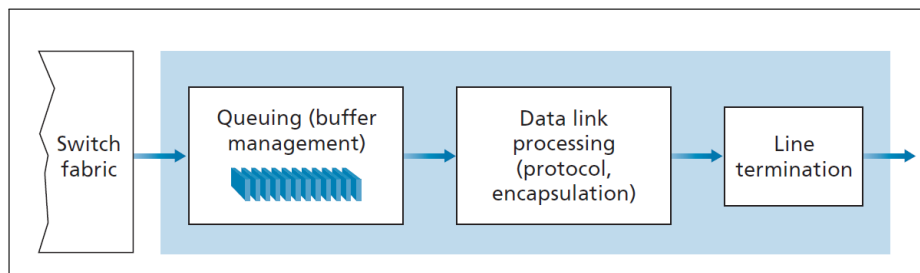


Figure 1: Output Port Processing

These operations constitute the final stage in the packet-forwarding pipeline of a router.

2 Where Does Queueing Occur?

Packet queues may form in two major locations inside a router:

1. **Input port queues**
2. **Output port queues**

The extent of queueing depends on:

- traffic arrival rates,
- switching fabric speed,
- link transmission rate.

When buffers overflow, packets are dropped — this is where packets are *actually* lost inside the Internet.

3 System Model

Assume:

- All input/output link speeds are equal: R_{line} packets/sec.
- There are N input ports and N output ports.
- All packets have fixed length.
- Arrivals are synchronous.

Let R_{switch} denote the switching fabric transfer rate.

Key Observation

If $R_{\text{switch}} = N \cdot R_{\text{line}}$, then input queueing is negligible because the fabric can clear N incoming packets before the next batch arrives.

4 Input Queueing and HOL Blocking

If the switching fabric is not fast enough relative to the input line rate, packets must join queues at the **input ports**. Even in crossbar switches with parallel transfer, inefficiencies can occur.

4.1 Head-of-the-Line (HOL) Blocking

HOL blocking occurs when:

- A packet at the front of an input queue is blocked (because its output port is busy).
- All packets behind it must wait—even if their output ports are free.

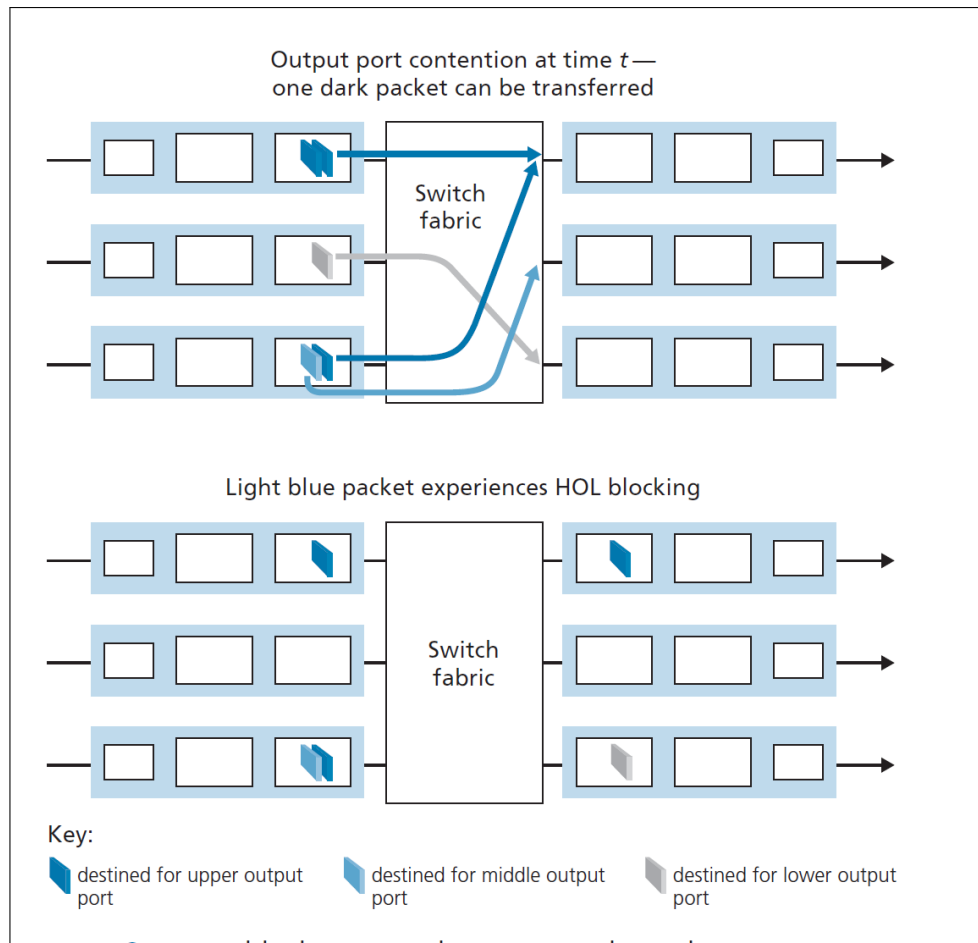


Figure 2: HOL Blocking at an Input-Queued Switch

Important Result

Input queues become unstable when the arrival rate reaches approximately

$$0.58 R_{\text{line}}$$

This leads to unbounded queue growth and packet loss.

5 Output Queueing

Even if $R_{\text{switch}} = N \cdot R_{\text{line}}$, output queues can still form.

Example scenario:

- All N input ports send packets to one output port.
- In the time needed to send one packet, N new packets may arrive.
- Thus, output queues grow rapidly.

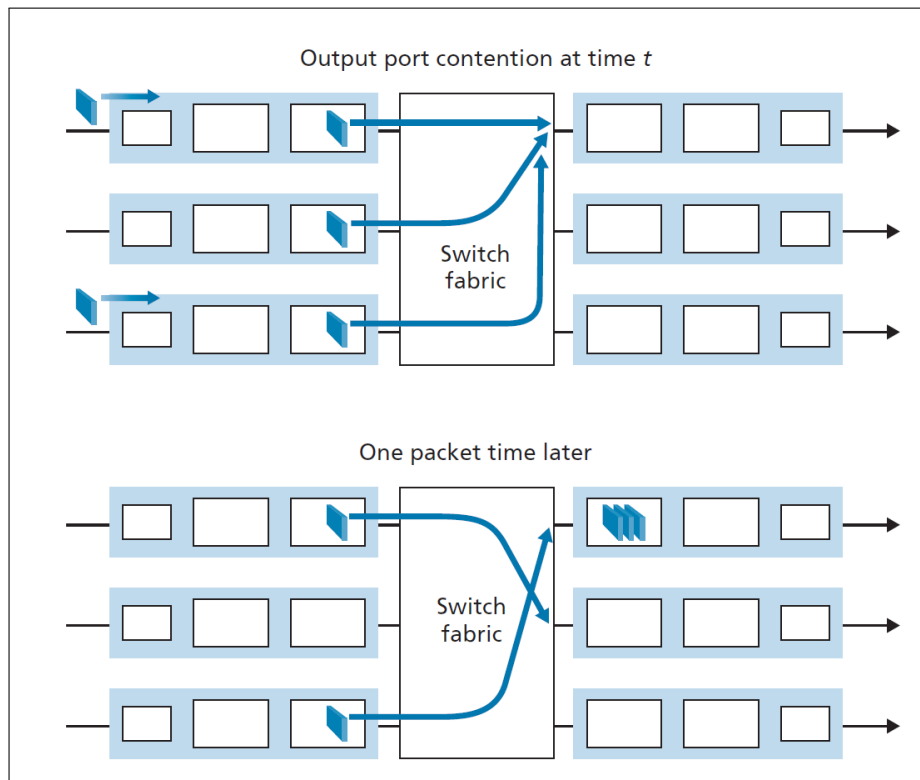


Figure 3: Output Port Queueing

If buffers fill up, packet loss occurs.

5.1 Drop Policies

When the output queue is full:

- **Drop-tail:** drop the arriving packet.
- **Queue replacement:** remove one or more older packets.
- **Early dropping:** signal congestion before the buffer becomes full.

6 Active Queue Management (AQM)

AQM techniques attempt to maintain smaller queue lengths and provide early congestion feedback.

Common AQM algorithms:

- **RED** (Random Early Detection)
- **PIE** (Proportional Integral controller Enhanced)
- **CoDel** (Controlled Delay)

These algorithms help reduce:

- queue buildup,
- latency,
- packet loss bursts,
- TCP global synchronization.